LAB SHEET 1

OBJECTIVE:

To design ER diagrams and convert entities and relationships into relational tables using proper schema notation.

- **Relational Schema**
  PART A – COLLEGE DATABASE
  STUDENT (USN, SName, Address, Phone, Gender)
  SEMSEC (SSID, Sem, Sec)
  CLASS (USN, SSID)
  SUBJECT (Subcode, Title, Sem, Credits)
  IAMARKS (USN, Subcode, SSID, Test1, Test2, Test3, FinalIA)

Entities and Attributes

1. **STUDENT**
   - USN (Primary Key)
   - SName
   - Address
   - Phone
   - Gender
2. **SEMSEC**
   - SSID (Primary Key)
   - Sem
   - Sec
3. **CLASS**
   - USN (Foreign Key)
   - SSID (Foreign Key)
4. **SUBJECT**
   - Subcode (Primary Key)
   - Title
   - Sem
   - Credits
5. **IAMARKS**

- USN (Foreign Key)
- Subcode (Foreign Key)
- SSID (Foreign Key)
- Test1
- Test2
- Test3
- FinalIA

DDL QUESTIONS

1. **Write SQL statements to create all the above relations with appropriate PRIMARY KEY and FOREIGN KEY constraints**

```sql
CREATE DATABASE DBMS_Parinita;
USE DBMS_Parinita;
CREATE TABLE STUDENT (
    USN VARCHAR(20) PRIMARY KEY,
    SName VARCHAR(50),
    Address VARCHAR(100),
    Phone VARCHAR(15),
    Gender VARCHAR(10)
);
CREATE TABLE SEMSEC (
    SSID INT PRIMARY KEY,
    Sem INT,
    Sec VARCHAR(5)
);
CREATE TABLE CLASS (
    USN VARCHAR(20),
    SSID INT,
    PRIMARY KEY (USN, SSID),
    FOREIGN KEY (USN) REFERENCES STUDENT(USN),
    FOREIGN KEY (SSID) REFERENCES SEMSEC(SSID)
);
```

```sql
CREATE TABLE SUBJECT (
    Subcode VARCHAR(20) PRIMARY KEY,
    Title VARCHAR(50),
    Sem INT,
    Credits INT
);
CREATE TABLE IAMARKS (
    USN VARCHAR(20),
    Subcode VARCHAR(20),
    SSID INT,
    Test1 INT,
    Test2 INT,
    Test3 INT,
    FinalIA INT,
    PRIMARY KEY (USN, Subcode, SSID),
    FOREIGN KEY (USN) REFERENCES STUDENT(USN),
    FOREIGN KEY (Subcode) REFERENCES SUBJECT(Subcode),
    FOREIGN KEY (SSID) REFERENCES SEMSEC(SSID)
);
```

PART B – COMPANY DATABASE

EMPLOYEE (SSN, Name, Address, Sex, Salary, Superssn, Dno)
DEPARTMENT (Dno, Dname, Mgrssn, Mgrstartdate)
DLOCATION (Dno, Dloc)
PROJECT (PNo, PName, PLocation, DNo)
WORKS_ON (SSN, PNo, Hours)

Entities and Attributes

1. **EMPLOYEE**
    - SSN (Primary Key)
    - Name
    - Address
    - Sex
    - Salary
    - Superssn (Foreign Key)
    - Dno (Foreign Key)
2. **DEPARTMENT**
    - Dno (Primary Key)
    - Dname

- Mgrssn
- Mgrstartdate

3. **DLOCATION**
    - Dno (Foreign Key)
    - Dloc
4. **PROJECT**
    - PNo (Primary Key)
    - PName
    - PLocation
    - DNo (Foreign Key)
5. **WORKS_ON**
    - SSN (Foreign Key)
    - PNo (Foreign Key)
    - Hours

SQL STATEMENTS

```sql
USE DBMS_Parinita;
CREATE TABLE EMPLOYEE (
    SSN INT PRIMARY KEY,
    Name VARCHAR(50),
    Address VARCHAR(100),
    Sex VARCHAR(10),
    Salary INT,
    Superssn INT,
    Dno INT
);
CREATE TABLE DEPARTMENT (
    Dno INT PRIMARY KEY,
    Dname VARCHAR(50),
    Mgrssn INT,
    Mgrstartdate DATE
);
CREATE TABLE DLOCATION (
    Dno INT,
    Dloc VARCHAR(50),
    PRIMARY KEY (Dno, Dloc),
    FOREIGN KEY (Dno) REFERENCES DEPARTMENT(Dno)
);
CREATE TABLE PROJECT (
    PNo INT PRIMARY KEY,
    PName VARCHAR(50),
    PLocation VARCHAR(50),
    DNo INT,
    FOREIGN KEY (DNo) REFERENCES DEPARTMENT(Dno)
);
```

```sql
CREATE TABLE WORKS_ON (
    SSN INT,
    PNo INT,
    Hours INT,
    PRIMARY KEY (SSN, PNo),
    FOREIGN KEY (SSN) REFERENCES EMPLOYEE(SSN),
    FOREIGN KEY (PNo) REFERENCES PROJECT(PNo)
);
ALTER TABLE EMPLOYEE
ADD FOREIGN KEY (Dno) REFERENCES DEPARTMENT(Dno);
ALTER TABLE EMPLOYEE
ADD FOREIGN KEY (Superssn) REFERENCES EMPLOYEE(SSN);
ALTER TABLE DEPARTMENT
ADD FOREIGN KEY (Mgrssn) REFERENCES EMPLOYEE(SSN);
```

PART C - CUSTOMER SALE SCHEMA

Database Schema for a customer-sale scenario

Customer(Cust id : integer, cust_name: string)

Item(item_id: integer, item_name:string, price: integer)

Sale(bill_no:integer,bill_data:date,**cust**_id: integer, item_id: integer, qty_sold: integer)

Entities

1. **CUSTOMER**
   - Cust_id (Primary Key)
   - cust_name
2. **ITEM**
   - item_id (Primary Key)
   - item_name
   - price
3. **SALE**
   - bill_no (Primary Key)
   - bill_date
   - cust_id (Foreign Key)
   - item_id (Foreign Key)
   - qty_sold

SQL STATEMENTS

```sql
CREATE DATABASE DBMS_Parinita;
USE DBMS_Parinita;
CREATE TABLE CUSTOMER (
    Cust_id INT PRIMARY KEY,
    Cust_name VARCHAR(50)
);
CREATE TABLE ITEM (
    Item_id INT PRIMARY KEY,
    Item_name VARCHAR(50),
    Price INT NOT NULL
);
CREATE TABLE SALE (
    Bill_no INT PRIMARY KEY,
    Bill_date DATE,
    Cust_id INT,
    Item_id INT,
    Qty_sold INT,
    FOREIGN KEY (Cust_id) REFERENCES CUSTOMER(Cust_id),
    FOREIGN KEY (Item_id) REFERENCES ITEM(Item_id)
);
```

DDL QUESTIONS

2. **Add a new column Email to the STUDENT relation**.

QUERY

```sql
CREATE DATABASE DBMS_Parinita;
USE DBMS_Parinita;
ALTER TABLE STUDENT
ADD Email VARCHAR(100);
DESC STUDENT;
```

OUTPUT

Result Grid | Filter Rows: | Export: | Wrap C

Field	Type	Null	Key	Default	Extra
USN	varchar(20)	NO	PRI	NULL	
SName	varchar(50)	YES		NULL	
Address	varchar(100)	YES		NULL	
Phone	varchar(15)	YES		NULL	
Gender	varchar(10)	YES		NULL	
Email	varchar(100)	YES		NULL	

3. **Apply NOT NULL constraints to Salary and Price attributes.**

QUERY

```
CREATE DATABASE DBMS_Parinita;
USE DBMS_Parinita;
ALTER TABLE EMPLOYEE
MODIFY Salary INT NOT NULL;
ALTER TABLE ITEM
MODIFY Price INT NOT NULL;
DESC EMPLOYEE;
DESC ITEM;
```

OUTPUT

Result Grid | Filter Rows: | Export: | Wrap C

Field	Type	Null	Key	Default	Extra
SSN	int(11)	NO	PRI	NULL	
Name	varchar(50)	YES		NULL	
Address	varchar(100)	YES		NULL	
Sex	varchar(10)	YES		NULL	
Salary	int(11)	NO		NULL	
Superssn	int(11)	YES	MUL	NULL	
Dno	int(11)	YES	MUL	NULL	

Result Grid | Filter Rows: | Export: | Wrap C

Field	Type	Null	Key	Default	Extra
Item_id	int(11)	NO	PRI	NULL	
Item_name	varchar(50)	YES		NULL	
Price	int(11)	NO		NULL	

4. **Rename attribute SName to StudentName in STUDENT relation.**

QUERY

```
CREATE DATABASE DBMS_Parinita;
USE DBMS_Parinita;
ALTER TABLE STUDENT
CHANGE SName StudentName VARCHAR(50);
DESC STUDENT;
```

OUTPUT

Result Grid | Filter Rows: | Export: | Wrap Cell

Field	Type	Null	Key	Default	Extra
USN	varchar(20)	NO	PRI	NULL	
StudentName	varchar(50)	YES		NULL	
Address	varchar(100)	YES		NULL	
Phone	varchar(15)	YES		NULL	
Gender	varchar(10)	YES		NULL	
Email	varchar(100)	YES		NULL	

5. **Drop the DLOCATION relation.**

QUERY

```
CREATE DATABASE DBMS_Parinita;
USE DBMS_Parinita;
DROP TABLE DLOCATION;
SHOW TABLES;
```

OUTPUT

There is no table named DLOCATION

Result Grid | Filter Rows:

Tables_in_dbms_parinita
customer
department
employee
iamarks
item
project
sale
semsec
student
subject
works_on

DML Questions (Data Manipulation Language):

1. **Insert at least 10 tuples into CUSTOMER, ITEM, and SALE relations.**

QUERY

```
CREATE DATABASE DBMS_Parinita;
USE DBMS_Parinita;
INSERT INTO CUSTOMER VALUES
(1,'Aman'),
(2,'Riya'),
(3,'Karan'),
(4,'Neha'),
(5,'Simran'),
(6,'Arjun'),
(7,'Priya'),
(8,'Rahul'),
(9,'Meena'),
(10,'Vikas');
```

```sql
INSERT INTO ITEM VALUES
(101,'Pen',20),
(102,'Notebook',50),
(103,'Bag',800),
(104,'Bottle',250),
(105,'Shoes',1200),
(106,'Tiffin',300),
(107,'Watch',1500),
(108,'Book',400),
(109,'Mouse',600),
(110,'Keyboard',900);
INSERT INTO SALE VALUES
(1,CURDATE(),1,103,1),
(2,CURDATE(),2,105,2),
(3,CURDATE(),3,101,5),
(4,CURDATE(),4,110,1),
(5,CURDATE(),5,107,1),
(6,CURDATE(),6,104,2),
(7,CURDATE(),7,108,3),
(8,CURDATE(),8,102,4),
(9,CURDATE(),9,109,1),
(10,CURDATE(),5,105,1);
```

OUTPUT

Result Grid | Filter Rows: | Edit:

Bill_no	Bill_date	Cust_id	Item_id	Qty_sold
1	2026-03-01	1	103	1
2	2026-03-01	2	105	2
3	2026-03-01	3	101	5
4	2026-03-01	4	110	1
5	2026-03-01	5	107	1
6	2026-03-01	6	104	2
7	2026-03-01	7	108	3
8	2026-03-01	8	102	4
9	2026-03-01	9	109	1
10	2026-03-01	5	105	1
NULL	NULL	NULL	NULL	NULL

2. **List all bills of the current date with customer names and item names.**

QUERY

```sql
CREATE DATABASE DBMS_Parinita;
USE DBMS_Parinita;
SELECT s.Bill_no,
       s.Bill_date,
       c.Cust_name,
       i.Item_name
FROM SALE s
JOIN CUSTOMER c ON s.Cust_id = c.Cust_id
JOIN ITEM i ON s.Item_id = i.Item_id
WHERE s.Bill_date = CURDATE();
```

OUTPUT

Result Grid | Filter Rows:

Bill_no	Bill_date	Cust_name	Item_name
1	2026-03-01	Aman	Bag
2	2026-03-01	Riya	Shoes
3	2026-03-01	Karan	Pen
4	2026-03-01	Neha	Keyboard
5	2026-03-01	Simran	Watch
6	2026-03-01	Arjun	Bottle
7	2026-03-01	Priya	Book
8	2026-03-01	Rahul	Notebook
9	2026-03-01	Meena	Mouse
10	2026-03-01	Simran	Shoes

3. **Display total bill amount using quantity and price.**

QUERY

```
CREATE DATABASE DBMS_Parinita;
USE DBMS_Parinita;
SELECT s.Bill_no,
       c.Cust_name,
       i.Item_name,
       s.Qty_sold,
       i.Price,
       (s.Qty_sold * i.Price) AS Total_Amount
FROM SALE s
JOIN CUSTOMER c ON s.Cust_id = c.Cust_id
JOIN ITEM i ON s.Item_id = i.Item_id;
```

OUTPUT

Result Grid | Filter Rows: | Export: | Wrap Cell Co

Bill_no	Cust_name	Item_name	Qty_sold	Price	Total_Amount
1	Aman	Bag	1	800	800
2	Riya	Shoes	2	1200	2400
3	Karan	Pen	5	20	100
4	Neha	Keyboard	1	900	900
5	Simran	Watch	1	1500	1500
6	Arjun	Bottle	2	250	500
7	Priya	Book	3	400	1200
8	Rahul	Notebook	4	50	200
9	Meena	Mouse	1	600	600
10	Simran	Shoes	1	1200	1200

4. **List customers who purchased products priced above 200.**

QUERY

```
CREATE DATABASE DBMS_Parinita;
USE DBMS_Parinita;
SELECT DISTINCT c.Cust_name
FROM SALE s
JOIN CUSTOMER c ON s.Cust_id = c.Cust_id
JOIN ITEM i ON s.Item_id = i.Item_id
WHERE i.Price > 200;
```

OUTPUT

Result Grid | Filter

Cust_name
Aman
Riya
Neha
Simran
Arjun
Priya
Meena

5. **Count number of products purchased by each customer**

QUERY

```
CREATE DATABASE DBMS_Parinita;
USE DBMS_Parinita;
SELECT c.Cust_name,
       COUNT(s.Item_id) AS Total_Products
FROM SALE s
JOIN CUSTOMER c ON s.Cust_id = c.Cust_id
GROUP BY c.Cust_name;
```

OUTPUT

Result Grid | Filter Rows:

Cust_name	Total_Products
Aman	1
Arjun	1
Karan	1
Meena	1
Neha	1
Priya	1
Rahul	1
Riya	1
Simran	2

6. **List products purchased by customer having Cust_id = 5.**

QUERY

```
CREATE DATABASE DBMS_Parinita;
USE DBMS_Parinita;
SELECT i.Item_name
FROM SALE s
JOIN ITEM i ON s.Item_id = i.Item_id
WHERE s.Cust_id = 5;
```

OUTPUT

Result Grid | Filter Ro

Item_name
Watch
Shoes

7. **Create a VIEW showing bill_no, bill_date, cust_id, item_id, price, qty,sold, amount.**

QUERY

```
CREATE DATABASE DBMS_Parinita;
USE DBMS_Parinita;
CREATE VIEW Bill_View AS
SELECT s.Bill_no,
       s.Bill_date,
       s.Cust_id,
       s.Item_id,
       i.Price,
       s.Qty_sold,
       (s.Qty_sold * i.Price) AS Amount
FROM SALE s
JOIN ITEM i ON s.Item_id = i.Item_id;
```

OUTPUT

Result Grid | Filter Rows: | Export: | Wrap Cell Co

Bill_no	Bill_date	Cust_id	Item_id	Price	Qty_sold	Amount
1	2026-03-01	1	103	800	1	800
2	2026-03-01	2	105	1200	2	2400
3	2026-03-01	3	101	20	5	100
4	2026-03-01	4	110	900	1	900
5	2026-03-01	5	107	1500	1	1500
6	2026-03-01	6	104	250	2	500
7	2026-03-01	7	108	400	3	1200
8	2026-03-01	8	102	50	4	200
9	2026-03-01	9	109	600	1	600
10	2026-03-01	5	105	1200	1	1200

8. **Create a VIEW showing daily sales date-wise for the last one week.**

QUERY

```
CREATE DATABASE DBMS_Parinita;
USE DBMS_Parinita;
CREATE VIEW Weekly_Sales AS
SELECT s.Bill_date,
       SUM(s.Qty_sold * i.Price) AS Total_Sales
FROM SALE s
JOIN ITEM i ON s.Item_id = i.Item_id
WHERE s.Bill_date >= CURDATE() - INTERVAL 7 DAY
GROUP BY s.Bill_date;
SELECT * FROM Weekly_Sales;
```

OUTPUT

Result Grid | Filter Rows:

Bill_date	Total_Sales
2026-03-01	9400